UNIFIED RISK ENGINE

From financial text to portfolio risk signals

An explainable offline-first prototype for wholesale banking

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DEMO PORTFOLIO • USD ~1B • REPLAY MODE

THE BUSINESS PROBLEM

Risk begins as text—and ends as exposure

News, filings and social chatter arrive in different formats. Analysts must decide what happened, which names are affected and whether it matters to the portfolio.

The prototype turns each item into a traceable signal, then routes it into two decision-support workflows.

Replay-first; live sources are optional

UNSTRUCTURED INPUT

01 **Normalize & link**

Clean text; resolve ticker aliases

↓

02 **Classify & score**

Sentiment • event • impact

↓

03 **Connect to risk**

Rebalance view + stress scenario

SYSTEM DESIGN

A replayable pipeline with two downstream paths



LIVE SOURCE UNAVAILABLE automatic fallback → seeded replay

DOWNSTREAM DECISION SUPPORT

01 Portfolio rebalancer

Constrained target weights, turnover and triggers

02 Portfolio stress test

Map event shocks to asset-class P&L and positions

IMPLEMENTATION HIGHLIGHTS

Small, inspectable modules keep the demo resilient

INGESTION

01 **Seeded crisis timeline**

02 RSS connectors; optional Finnhub

03 Replay fallback on live failure

Why: repeatable demos without credentials.

RISK ENGINE

01 **Text cleaning + ticker aliases**

02 Sentiment and event classification

03 Impact score with rationale

Why: expose what drove each signal.

DECISION SUPPORT

01 **Rebalancer with weight/turnover limits**

02 Multi-asset stress simulation

03 FastAPI + live event stream

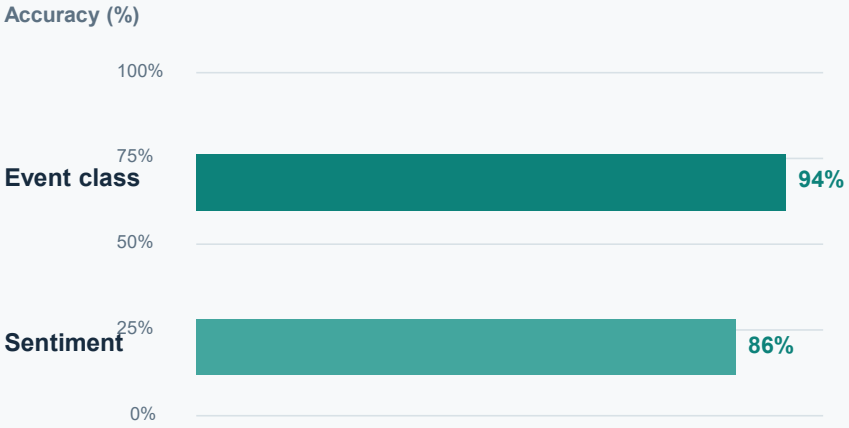
Why: connect text to portfolio context.

CPU-friendly • local cache • configurable universe

KEY RESULTS / SEED RUN

Measured outputs from the reproducible demo

Seed-set classification



n = 50 synthetic hand-labeled documents

Portfolio workflows

15-name rebalance universe

5.49%

observed turnover • 25% configured cap

$1.0B

synthetic portfolio • 150 positions

−$39.30M

modeled P&L under synthetic geopolitical shock

Important: seed metrics are not an independent benchmark; lexicon/rules were tuned on this demo set. No measured efficiency gain.

Make the path from headline to exposure visible

SIMULATED BREAKING HEADLINE

“Strait of Hormuz transit interruption triggers oil shock and broad market risk.”

Geopolitical | Negative sentiment | XOM + CVX linked

IMPACT SCORE

7.84 / 10

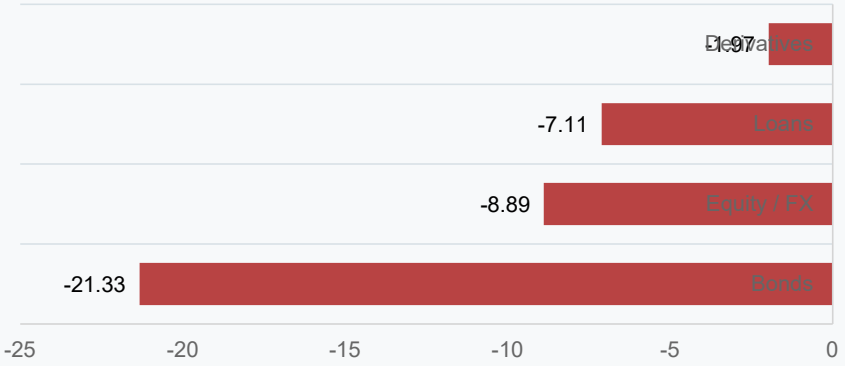
SCENARIO SHOCKS

Oil +28% • Equities −11.2% • Spreads +168 bp

PORTFOLIO VALUE

$1.000B → $960.70M

Illustrative modeled P&L by asset class ($M)



Illustrative sensitivity model—not a forecast, valuation or loss estimate. Outputs depend on synthetic exposures and shock assumptions.

LIMITATIONS & NEXT STEPS

Prototype boundaries—and the path to production

LIMITATIONS

- Synthetic replay, portfolio and returns; no real transaction sample was supplied.
- 50 synthetic labels only; no independent benchmark or out-of-sample test.
- Rules/lexicon and simplified scenario valuation do not capture full market dynamics.
- No quantified analyst-time or efficiency benchmark.

NEXT STEPS

01 Evaluate on licensed, independently labeled financial text.

02 Connect approved news/social and market data with cache + provenance.

03 Replace synthetic positions with mapped real portfolio data and bank-approved valuation models.

04 Add analyst review, access controls, monitoring and measured workflow-time baseline.

Current role: triage + explainability prototype • not automated investment or credit advice